12.3 Normal Operators

Section 2 collected everything a self-adjoint operator gives us: real eigenvalues, orthogonal eigenspaces, an invariant orthogonal complement. That is exactly the equipment an orthonormal basis of eigenvectors needs. But over $\mathbb{C}$ the hypothesis is too heavy, and this section finds the lighter one that still carries the load. It is a condition already met in Chapter 11, and the rest of the chapter runs on it.

As throughout, F is $\mathbb{R}$ or $\mathbb{C}$, inner products are linear in the first slot, and every inner product space is finite-dimensional unless we say otherwise.

12.3.1 The hypothesis that self-adjointness overshoots

Take the operator on $\mathbb{C}^2$ whose matrix in the standard basis is $\mathbf{D} = \operatorname{diag}(1, i)$. The standard basis is orthonormal and consists of eigenvectors of $\mathbf{D}$, so this operator has everything the spectral theorem promises. Yet it is not self-adjoint: one of its eigenvalues is i, and a self-adjoint operator has only real eigenvalues (Theorem 12.2.1). So over $\mathbb{C}$ self-adjointness cannot be the condition we are looking for. It is sufficient and not necessary, and we are throwing away every operator with a nonreal eigenvalue — including every unitary operator other than the trivial ones.

What survives? Compare the two computations. For a self-adjoint T we have $T^*T = TT = TT^*$. For $\mathbf{D} = \operatorname{diag}(1, i)$ we have $\mathbf{D}^* = \operatorname{diag}(1, -i)$, which is a different matrix, but

$$\mathbf{D}^*\mathbf{D} = \operatorname{diag}(1, 1) = \mathbf{D}\mathbf{D}^*.$$

Neither operator equals its adjoint; both **commute** with their adjoints. That is the common feature, and Chapter 11 already gave it a name.

A normal operator is one that never gets in its adjoint's way: the two may be applied in either order.

Recall Definition 11.6.3: an operator $T \in \mathcal{L}(V)$ is **normal** when $T^*T = TT^*$. That is a single equation between two operators, with no vectors and no quantifiers in it, and it is strictly weaker than $T^* = T$. Section 6 of Chapter 11 checked the definition on self-adjoint operators, on unitary ones, on $i\mathbf{I}$, and on diagonal matrices, and found the 2×2 Jordan block $\mathbf{J}_2(0)$ failing it. We take those for granted and go one step further: normality is best understood as a statement about **lengths**, and that is the content of the first theorem below.

Before it, a wider catalog, since the class is larger than Chapter 11's four examples suggest.

Example. Which of these are normal?

Decide which of the following matrices are normal, with the standard inner product on $\mathbb{C}^n$ or $\mathbb{R}^n$. For those that are, say which familiar family they belong to.

(a) $\mathbf{A} = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$ over $\mathbb{C}$.

(b) $\mathbf{B} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ over $\mathbb{R}$.

(c) $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ over $\mathbb{C}$.

(d) $\mathbf{C} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$ over $\mathbb{C}$.

(e) $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$ over $\mathbb{R}$.

(f) $\mathbf{N} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ over $\mathbb{R}$.

Solution.

(a) Normal. $\mathbf{A}^* = \mathbf{A}$, since transposing swaps the off-diagonal entries and conjugating swaps them back, so $\mathbf{A}$ is Hermitian and both products equal $\mathbf{A}^2$. Every **self-adjoint** operator is normal.

(b) Normal. $\mathbf{B}^\top = -\mathbf{B}$, so $\mathbf{B}^\top\mathbf{B} = -\mathbf{B}^2 = \mathbf{B}\mathbf{B}^\top$. An operator with $T^* = -T$ is called **skew-adjoint**, and the same one-line computation makes every such operator normal.

(c) Normal. $\mathbf{U}^*\mathbf{U} = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \mathbf{I}$, so $\mathbf{U}$ is **unitary** (Definition 11.7.2), and then $\mathbf{U}^*\mathbf{U} = \mathbf{I} = \mathbf{U}\mathbf{U}^*$.

(d) Normal. Each row of $\mathbf{C}$ is the previous one shifted cyclically to the right; such a matrix is called a **circulant**. Writing out the products,

$$\mathbf{C}^\top\mathbf{C} = \mathbf{C}\mathbf{C}^\top = \begin{pmatrix} 14 & 11 & 11 \\ 11 & 14 & 11 \\ 11 & 11 & 14 \end{pmatrix}.$$

Both entries in position (i, j) are the dot product of two rows of $\mathbf{C}$, or of two columns, and cyclic shifting makes those two lists of dot products agree. Section 9 explains circulants properly and diagonalizes all of them at once.

(e) Normal, and in none of the families above: $\mathbf{M}^\top \neq \pm\mathbf{M}$ and $\mathbf{M}^\top\mathbf{M} = 5\mathbf{I} \neq \mathbf{I}$. Still,

$$\mathbf{M}^\top\mathbf{M} = \mathbf{M}\mathbf{M}^\top = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}.$$

This is $\sqrt{5}$ times a rotation of the plane, which is the rotation-scaling block $\mathbf{\Lambda}(2 + i)$ transposed (Definition 10.4.1). Normality is genuinely wider than "self-adjoint or unitary".

(f) **Not** normal:

$$\mathbf{N}^\top\mathbf{N} = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}, \qquad \mathbf{N}\mathbf{N}^\top = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}.$$

The $(1, 1)$ entries already disagree, $1 \neq 2$. This matrix is diagonalizable, with eigenvalues 1 and 2; Section 4 shows what it costs to be diagonalizable and not normal.

12.3.2 Normality is a statement about lengths

The definition compares two operators. The next theorem turns it into a comparison of two numbers, one vector at a time, and that is the form every later proof uses. The engine is the

zero test of Section 2: a self-adjoint operator S with $\langle Sv, v\rangle = 0$ for all v is zero.

Theorem 12.3.1. Characterizations of Normality

Let V be a finite-dimensional inner product space over F and let $T \in \mathcal{L}(V)$. The following are equivalent.

(a) T is normal.
(b) $\|Tv\| = \|T^*v\|$ for every $v \in V$.
(c) $\langle Tu, Tv\rangle = \langle T^*u, T^*v\rangle$ for all $u, v \in V$.

Suppose in addition that $F = \mathbb{C}$, and set

$$A = \tfrac{1}{2}(T + T^*), \qquad B = \tfrac{1}{2i}(T - T^*),$$

so that A and B are self-adjoint and $T = A + iB$. Then (a)–(c) are further equivalent to

(d) $AB = BA$.

Idea. Condition (a) says that the operator $S = T^*T - TT^*$ is zero. That operator is self-adjoint, so by the zero test of Theorem 12.2.5 it is enough to know $\langle Sv, v\rangle = 0$ for every v — and $\langle Sv, v\rangle$ is exactly $\|Tv\|^2 - \|T^*v\|^2$. So (a) and (b) are the same statement read at two different levels of resolution. Condition (c) is (b) with the two slots unglued, and it costs nothing in one direction and a specialization in the other. Condition (d) is the operator version of writing a complex number as $a+bi$; that decomposition appeared in Exercise (C2: Skew-adjoint operators) (c), and multiplying out $TT^* - T^*T$ in terms of A and B leaves only the commutator.

Proof.

(a) $\Rightarrow$ (c). For all $u, v \in V$, by Definition 11.6.1 and $T^{**} = T$ (Theorem 11.6.2 (d)),

$$\langle Tu, Tv\rangle = \langle T^*Tu, v\rangle, \qquad \langle T^*u, T^*v\rangle = \langle TT^*u, v\rangle.$$

Since $T^*T = TT^*$, the two right-hand sides agree.

(b) $\Rightarrow$ (b). Take $u = v$. Then $\|Tv\|^2 = \|T^*v\|^2$, and both norms are nonnegative reals, so they are equal.

(c) $\Rightarrow$ (a). Put $S = T^*T - TT^*$. Then S is self-adjoint: by Theorem 11.6.2 (a), (c), and (d),

$$S^* = (T^*T)^* - (TT^*)^* = T^*T^{**} - T^{**}T^* = T^*T - TT^* = S.$$

For any $v \in V$, using Definition 11.6.1 twice,

$$\begin{aligned}\langle Sv, v\rangle &= \langle T^*Tv, v\rangle - \langle TT^*v, v\rangle \\ &= \langle Tv, Tv\rangle - \langle T^*v, T^*v\rangle \\ &= \|Tv\|^2 - \|T^*v\|^2 = 0\end{aligned}$$

by (b). Since S is self-adjoint, Theorem 12.2.5 gives $S = 0$, that is $T^*T = TT^*$.

Now let $F = \mathbb{C}$. By Theorem 11.6.2 (a), (b), and (d),

$$A^* = \tfrac{1}{2}(T^* + T) = A,$$
$$B^* = \overline{\left(\tfrac{1}{2i}\right)}(T^* - T) = \tfrac{1}{2i}(T - T^*) = B,$$

so both are self-adjoint, and $A + iB = \frac{1}{2}(T + T^*) + \frac{1}{2}(T - T^*) = T$; this is the decomposition of Exercise (C2: Skew-adjoint operators) (c). Expanding $T = A + iB$ and $T^* = A - iB$,

$$TT^* = (A + iB)(A - iB) = A^2 + B^2 + i(BA - AB),$$
$$T^*T = (A - iB)(A + iB) = A^2 + B^2 + i(AB - BA).$$

Subtracting, $TT^* - T^*T = 2i(BA - AB)$. So T is normal exactly when $AB = BA$, which is (a) $\Leftrightarrow$ (d). This proves the theorem.

■

Condition (b) is the one to remember. It says that T and T^* stretch every single vector by the same amount, even though they need not move it in the same direction. Everything in the next three sections is squeezed out of that sentence. Condition (d) records that normality over $\mathbb{C}$ is the operator analogue of the fact that any two real numbers commute: split T into a "real part" and an "imaginary part", and normality is precisely the failure of those two parts to interfere.

Quick check. The symmetric matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and the skew-symmetric matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ are both normal, the first being self-adjoint and the second skew-adjoint. Is their sum normal?

Solution. No. The sum is $\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = 2\mathbf{J}_2(0)$, which is twice the standard nonnormal example: it kills $\mathbf{e}_1$ and sends $\mathbf{e}_2$ to a vector of length 2, while its transpose does the opposite, so condition (b) fails at $\mathbf{v} = \mathbf{e}_1$. The normal operators do not form a subspace of $\mathcal{L}(V)$, and they are not closed under composition either.

12.3.3 What normality buys

Four consequences, all short, all obtained by turning some statement about T into the same statement about T^*.

Corollary 12.3.2. Kernel, Image, and an Orthogonal Splitting

Let $T \in \mathcal{L}(V)$ be normal on a finite-dimensional inner product space V. Then:

(a) $\ker T = \ker T^*$;

(b) $\operatorname{im} T = \operatorname{im} T^*$;

(c) $V = \ker T \oplus \operatorname{im} T$, and the two summands are orthogonal to each other.

Proof.

(a) For $v \in V$, Theorem 12.3.1 (b) gives $\|Tv\| = \|T^*v\|$. So $Tv = 0$ if and only if $\|Tv\| = 0$, if and only if $\|T^*v\| = 0$, if and only if $T^*v = 0$.

(b) By Theorem 11.6.5 (b) applied to T^* in place of T, and using $T^{**} = T$ (Theorem 11.6.2 (d)),

$$\operatorname{im} T = \operatorname{im} T^{**} = (\ker T^*)^\perp = (\ker T)^\perp = \operatorname{im} T^*,$$

where the third equality is (a) and the fourth is Theorem 11.6.5 (b) for T.

(c) By (b), $\operatorname{im} T = (\ker T)^\perp$, and $V = \ker T \oplus (\ker T)^\perp$ by Theorem 11.3.2 (a). Orthogonality of the summands is the definition of the orthogonal complement.

■

Part (c) fails badly without normality, and the failure is not subtle: for a general operator $\ker T$ and $\operatorname{im} T$ can be **equal**. Take T with matrix $\mathbf{J}_2(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ on $\mathbb{C}^2$. Then $T\mathbf{e}_1 = \mathbf{0}$ and $T\mathbf{e}_2 = \mathbf{e}_1$, so

$$\ker T = \operatorname{span}(\mathbf{e}_1) = \operatorname{im} T.$$

The sum $\ker T + \operatorname{im} T$ is one-dimensional, so it is neither direct nor all of $\mathbb{C}^2$. Normality is exactly what rules this out.

Theorem 12.3.3. An Eigenvector of a Normal Operator Is an Eigenvector of Its Adjoint

Let $T \in \mathcal{L}(V)$ be normal, let $\lambda \in F$, and let $v \in V$. Then

$$Tv = \lambda v \quad \text{if and only if} \quad T^*v = \overline{\lambda}\, v.$$

In particular $E_\lambda(T) = E_{\overline{\lambda}}(T^*)$, and $\lambda \in \operatorname{spec}(T)$ if and only if $\overline{\lambda} \in \operatorname{spec}(T^*)$.

Idea. Both sides say that v lies in a kernel: the left in $\ker(T - \lambda \operatorname{id}_V)$, the right in $\ker(T^* - \overline{\lambda} \operatorname{id}_V)$. Those two operators are adjoint to each other, so Corollary 12.3.2 (a) closes the gap — provided $T - \lambda \operatorname{id}_V$ is itself normal, which is the only thing to check.

Proof. Write $S = T - \lambda \operatorname{id}_V$. By Theorem 11.6.2 (a), (b), and (d), $S^* = T^* - \overline{\lambda} \operatorname{id}_V$. Since $T^*T = TT^*$ and id_V commutes with everything,

$$\begin{aligned} S^*S &= T^*T - \overline{\lambda}T - \lambda T^* + |\lambda|^2 \operatorname{id}_V \\ &= TT^* - \lambda T^* - \overline{\lambda}T + |\lambda|^2 \operatorname{id}_V = SS^*, \end{aligned}$$

so S is normal. By Corollary 12.3.2 (a), $\ker S = \ker S^*$. Now $Tv = \lambda v$ says $v \in \ker S$, and $T^*v = \overline{\lambda}v$ says $v \in \ker S^*$, so the two conditions agree. The statement about eigenspaces is the equality $\ker S = \ker S^*$ rewritten, and the statement about spectra follows because an eigenspace is nonzero exactly when the scalar is an eigenvalue (Theorem 9.2.1).

■

The theorem is the reason normal operators are so easy to handle: T and T^* have the *same* eigenvectors, with conjugate eigenvalues. The proof of the next corollary is the classical self-adjoint argument (Theorem 12.2.3) with one conjugate bar inserted, and that bar is exactly what Theorem 12.3.3 supplies.

Corollary 12.3.4. Eigenvectors for Distinct Eigenvalues Are Orthogonal

Let $T \in \mathcal{L}(V)$ be normal, let $\lambda \neq \mu$ be eigenvalues of T, and let $Tu = \lambda u$ and $Tv = \mu v$. Then $\langle u, v\rangle = 0$. Hence $E_\lambda(T) \perp E_\mu(T)$ whenever $\lambda \neq \mu$.

Proof. Compute $\langle Tu, v\rangle$ in two ways. First, $\langle Tu, v\rangle = \langle \lambda u, v\rangle = \lambda\langle u, v\rangle$. Second, by Definition 11.6.1 and then Theorem 12.3.3 applied to v,

$$\langle Tu, v\rangle = \langle u, T^*v\rangle = \langle u, \overline{\mu}\, v\rangle = \mu\langle u, v\rangle,$$

the last step because a scalar leaves the second slot conjugated. Subtracting, $(\lambda-\mu)\langle u, v\rangle = 0$, and $\lambda - \mu \neq 0$, so $\langle u, v\rangle = 0$. Every vector of $E_\lambda(T)$ is orthogonal to every vector of $E_\mu(T)$ by the same computation.

■

Theorem 12.3.5. A Normal Operator Has No Hidden Kernel

Let $T \in \mathcal{L}(V)$ be normal. Then $\ker T^k = \ker T$ for every integer $k \geq 1$. In particular, if $T^k = 0$ for some $k \geq 1$ — that is, if T is **nilpotent** (Definition 10.2.1) — then $T = 0$.

Idea. The inclusion $\ker T \subseteq \ker T^2$ is free, and the whole content is the reverse one. If $T^2v = 0$, then the vector $w = Tv$ lies in $\ker T$, hence also in $\ker T^*$ by Corollary 12.3.2; and a vector that is killed by T^* and is itself in the image of T must be 0, because pairing it with itself moves T across the inner product and hits the T^*.

Proof. First take $k = 2$. The inclusion $\ker T \subseteq \ker T^2$ holds for every operator. Conversely, suppose $T^2v = 0$ and put $w = Tv$. Then $Tw = 0$, so $T^*w = 0$ by Corollary 12.3.2 (a). Therefore, by Definition 11.6.1,

$$\begin{aligned}\|w\|^2 = \langle w, w\rangle &= \langle Tv, w\rangle \\ &= \langle v, T^*w\rangle = \langle v, 0\rangle = 0,\end{aligned}$$

so $w = Tv = 0$ and $v \in \ker T$. Hence $\ker T^2 = \ker T$.

Now induct on k. The cases $k = 1, 2$ are done. Let $k \geq 2$ and assume $\ker T^k = \ker T$. If $T^{k+1}v = 0$, then $T^2(T^{k-1}v) = 0$, so $T^kv = T(T^{k-1}v) = 0$ by the case $k = 2$, and then $Tv = 0$ by the inductive hypothesis. So $\ker T^{k+1} \subseteq \ker T$, and the reverse inclusion is again free.

Finally, if $T^k = 0$ for some $k \geq 1$, then $\ker T = \ker T^k = V$, so $T = 0$.

■

This is the sentence that explains why normal operators never produce Jordan blocks. Over $\mathbb{C}$, a Jordan block of size $m \geq 2$ for the eigenvalue λ contributes a vector killed by $(T - \lambda \operatorname{id}_V)^m$ but not by $T - \lambda \operatorname{id}_V$ (Definition 10.2.2). For a normal T the operator $T - \lambda \operatorname{id}_V$ is normal, as the proof of Theorem 12.3.3 showed, so Theorem 12.3.5 forbids such a vector. Every block in the Jordan form of a normal operator therefore has size 1, and a normal operator on a complex space is diagonalizable (Theorem 10.3.1, Definition 9.4.1).

Remark. That is diagonalizable, not yet **orthogonally** diagonalizable. The eigenspaces of distinct eigenvalues are already mutually orthogonal by Corollary 12.3.4, so all that is missing is an orthonormal basis inside each eigenspace, which Gram–Schmidt supplies (Theorem 11.2.2). Section 4 proves the whole statement in one stroke instead, from Schur's theorem, because that proof also delivers the converse and works without Chapter 10.

Two converses are worth flagging now, so that you know they are coming and do not try to prove them here. A normal operator whose eigenvalues are all real is self-adjoint, and a normal operator whose eigenvalues all have modulus 1 is unitary. Both are false without normality, and neither can be reached with the tools of this section: they need an orthonormal basis of eigenvectors, which is the business of Section 4.

Before the warning, one structural fact deserves to be stated where later sections can lean on it. It is the reason a normal operator can be taken apart orthogonally at all.

Proposition 12.3.6. An Invariant Subspace of a Normal Operator Is Reducing

Let $T \in \mathcal{L}(V)$ be normal on a finite-dimensional inner product space, and let $U \subseteq V$ be T-invariant. Then U is T^*-invariant, $U^\perp$ is T-invariant, and $T|_U$ is a normal operator on U.

Idea. Write the matrix in an orthonormal basis adapted to U. Invariance kills the block below the diagonal, and normality, read on the top-left block alone, kills the block above it. The trace is what converts "these two blocks agree" into "that block is zero".

Proof. Choose an orthonormal basis $\mathcal{B}_U = (e_1, \dots, e_r)$ of U and extend it to an orthonormal basis $\mathcal{B}$ of V (Corollary 11.2.3). Since $Te_j \in U$ for $j \leq r$, the first r columns of $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$ have zeros below row r, so

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{O} & \mathbf{D} \end{pmatrix},$$

with $\mathbf{B}$ of size $r \times r$. Because $\mathcal{B}$ is orthonormal, $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{A}^*$ (Theorem 11.6.3), and T normal means $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$. Comparing the $(1,1)$ blocks (Theorem 8.1.1),

$$(\mathbf{A}^*\mathbf{A})_{11} = \mathbf{B}^*\mathbf{B}, \qquad (\mathbf{A}\mathbf{A}^*)_{11} = \mathbf{B}\mathbf{B}^* + \mathbf{C}\mathbf{C}^*.$$

Taking traces and using $\operatorname{tr}(\mathbf{B}^*\mathbf{B}) = \operatorname{tr}(\mathbf{B}\mathbf{B}^*)$ (Theorem 0.9.6 (3)) gives $\operatorname{tr}(\mathbf{C}\mathbf{C}^*) = 0$. But $\operatorname{tr}(\mathbf{C}\mathbf{C}^*) = \sum_{i,j} |c_{ij}|^2$, a sum of nonnegative reals, so $\mathbf{C} = \mathbf{O}$.

Hence $\mathbf{A} = \mathbf{B} \oplus \mathbf{D}$. Reading the last $n - r$ columns, $Te_j \in U^\perp$ for $j > r$, so $U^\perp$ is T-invariant; reading the first r columns of $\mathbf{A}^*$, U is T^*-invariant. Finally $[T|_U]_{\mathcal{B}_U}^{\mathcal{B}_U} = \mathbf{B}$ and the $(1,1)$ blocks above now read $\mathbf{B}^*\mathbf{B} = \mathbf{B}\mathbf{B}^*$, so $T|_U$ is normal. This shows all three claims.

■

Warning. **Normality cannot be read off a corner.** A principal submatrix of a normal matrix need not be normal. The 3×3 cyclic shift

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

is a permutation matrix, hence orthogonal (Exercise (B3: Permutations are orthogonal)), hence normal; but deleting its last row and column leaves $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, the transpose of $\mathbf{J}_2(0)$, which is not normal. Equivalently: the operator that $\mathbf{S}$ compresses to $\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2)$ is not normal, because that subspace is not $\mathbf{S}$-invariant. What **is** true is that an invariant subspace of a normal operator is automatically invariant under the adjoint too, and the restriction there is normal; that is Proposition 12.3.6, proved just above.

12.3.4 Exercises

12.3.4.1 A. Check your understanding

Exercise. A1

(a) State the definition of a normal operator, and state one equivalent condition phrased in terms of norms.

(b) True or false: every self-adjoint operator is normal, and every normal operator is self-adjoint. Justify both halves.

(c) Let T be normal with $Tv = 3v$. What is T^*v? What if $Tv = 3iv$?

(d) Give a normal operator on $\mathbb{R}^2$ that is neither self-adjoint nor unitary.

(e) True or false: if S and T are normal then $S + T$ is normal. Justify your answer.

(f) Explain in one sentence why a normal operator cannot be nilpotent unless it is zero.

Solution.

(a) T is normal when $T^*T = TT^*$ (Definition 11.6.3). Equivalently, $\|Tv\| = \|T^*v\|$ for every $v \in V$ (Theorem 12.3.1 (b)).

(b) The first half is true: if $T^* = T$ then both products are T^2. The second half is false: $\operatorname{diag}(1, i)$ on $\mathbb{C}^2$ is normal and has a nonreal eigenvalue, so it is not self-adjoint (Theorem 12.2.1).

(c) $T^*v = 3v$ in the first case and $T^*v = -3iv$ in the second, by Theorem 12.3.3: the adjoint has the conjugate eigenvalue on the same eigenvector.

(d) $\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$ works (Example (Which of these are normal?) (e)): it is not symmetric, and $\mathbf{M}^\top \mathbf{M} = 5\mathbf{I} \neq \mathbf{I}$. Any $\mathbf{\Lambda}(\lambda)$ with $|\lambda| \neq 1$ and $\operatorname{Im} \lambda \neq 0$ does.

(e) False. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ is a sum of two normal matrices and is not normal.

(f) If $T^k = 0$ then $\ker T = \ker T^k = V$ by Theorem 12.3.5, so $T = 0$.

12.3.4.2 B. Practice

Exercise. B1: Determine which are normal

Determine which of the following matrices are normal, with the standard inner product. Justify your answer; for those that are not, exhibit a vector $\mathbf{v}$ with $\|\mathbf{Av}\| \neq \|\mathbf{A}^*\mathbf{v}\|$.

(a) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ over $\mathbb{R}$.

(b) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ over $\mathbb{R}$.

(c) $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ over $\mathbb{C}$.

(d) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$ over $\mathbb{R}$.

(e) $\begin{pmatrix} i & 1 \\ 0 & i \end{pmatrix}$ over $\mathbb{C}$.

Solution.

(a) Normal: the matrix is symmetric, so it is self-adjoint over $\mathbb{R}$, and self-adjoint implies normal.

(b) Not normal. $\mathbf{A}^\top\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. With $\mathbf{v} = \mathbf{e}_1$, $\|\mathbf{Ae}_1\| = \|(1,0)\| = 1$ while $\|\mathbf{A}^\top\mathbf{e}_1\| = \|(1,1)\| = \sqrt{2}$.

(c) Normal. $\mathbf{A}^* = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -\mathbf{A}$, so $\mathbf{A}$ is skew-adjoint and $\mathbf{A}^*\mathbf{A} = -\mathbf{A}^2 = \mathbf{A}\mathbf{A}^*$. (Concretely both products are $\mathbf{I}$.)

(d) Normal: the matrix is $(1) \oplus \mathbf{\Lambda}(i)$, a direct sum of a 1×1 block and a rotation by a quarter turn, and its columns are orthonormal, so it is orthogonal (Theorem 11.7.3 (f)).

(e) Not normal. $\mathbf{A}^*\mathbf{A} = \begin{pmatrix} 1 & -i \\ i & 2 \end{pmatrix}$ and $\mathbf{A}\mathbf{A}^* = \begin{pmatrix} 2 & -i \\ i & 1 \end{pmatrix}$; with $\mathbf{v} = \mathbf{e}_1$ we get $\|\mathbf{Ae}_1\| = |i| = 1$ and $\|\mathbf{A}^*\mathbf{e}_1\| = \|(-i,1)\| = \sqrt{2}$. Adding $i\mathbf{I}$ to a nonnormal matrix does not repair it: $\mathbf{A} = i\mathbf{I} + \mathbf{J}_2(0)$ and $i\mathbf{I}$ is normal, but sums of normal matrices need not be normal.

Exercise. B2: The splitting of a singular normal operator

Let $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{R})$, and let $T = T_{\mathbf{C}}$ be the operator it defines on $\mathbb{R}^3$.

(a) Verify that T is normal.

(b) Compute $\ker T$ and $\ker T^*$ and check that they agree.

(c) Compute $\operatorname{im} T$, and verify directly that $\mathbb{R}^3 = \ker T \oplus \operatorname{im} T$ with the two summands orthogonal, as Corollary 12.3.2 (c) predicts.

Solution.

(a) Each row of $\mathbf{C}$ is the previous one shifted cyclically, so $\mathbf{C}$ is a circulant. Explicitly,

$$\mathbf{C}^\top \mathbf{C} = \mathbf{C}\mathbf{C}^\top = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix},$$

so T is normal.

(b) $\mathbf{C}\mathbf{x} = \mathbf{0}$ reads $x_1 = x_2 = x_3$, because the first two rows give $x_1 = x_2$ and $x_2 = x_3$. So $\ker T = \operatorname{span}((1,1,1))$. The transpose $\mathbf{C}^\top$ has rows $(1,0,-1)$, $(-1,1,0)$, $(0,-1,1)$, and $\mathbf{C}^\top \mathbf{x} = \mathbf{0}$ again reads $x_1 = x_3$ and $x_1 = x_2$. The two kernels agree, as Corollary 12.3.2 (a) requires.

(c) The rank is $3 - 1 = 2$ by Theorem 2.3.1, and the first two columns $(1,0,-1)$ and $(-1,1,0)$ are independent, so

$$\operatorname{im} T = \operatorname{span}\big((1,0,-1),\ (-1,1,0)\big).$$

Both spanning vectors have coordinate sum 0, hence are orthogonal to $(1,1,1)$, so $\operatorname{im} T \perp \ker T$. Since $\dim \ker T + \dim \operatorname{im} T = 1 + 2 = 3$ and the two subspaces meet only in $\mathbf{0}$ (being orthogonal), $\mathbb{R}^3 = \ker T \oplus \operatorname{im} T$. Indeed $\operatorname{im} T = \{\mathbf{x} : x_1 + x_2 + x_3 = 0\} = (\ker T)^\perp$.

Exercise. B3: Triangular and normal, by hand

Let $\mathbf{A} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in M_2(\mathbb{C})$.

(a) Compute the $(1,1)$ entries of $\mathbf{A}^*\mathbf{A}$ and of $\mathbf{A}\mathbf{A}^*$.

(b) Hence prove that $\mathbf{A}$ is normal if and only if $b = 0$, that is, if and only if $\mathbf{A}$ is diagonal.

(c) Which upper triangular matrices in $M_2(\mathbb{C})$ are unitary?

Hint: for (a), you only need one entry of each product.

Solution.

(a) The $(1,1)$ entry of $\mathbf{A}^*\mathbf{A}$ is the squared norm of the first **column** of $\mathbf{A}$, namely $|a|^2$. The $(1,1)$ entry of $\mathbf{A}\mathbf{A}^*$ is the squared norm of the first **row**, namely $|a|^2 + |b|^2$. In

full,

$$\mathbf{A}^*\mathbf{A} = \begin{pmatrix} |a|^2 & \overline{a}b \\ a\overline{b} & |b|^2 + |d|^2 \end{pmatrix},$$

$$\mathbf{A}\mathbf{A}^* = \begin{pmatrix} |a|^2 + |b|^2 & b\overline{d} \\ \overline{b}d & |d|^2 \end{pmatrix}.$$

(b) ($\Rightarrow$) If $\mathbf{A}$ is normal, comparing the $(1,1)$ entries gives $|a|^2 = |a|^2 + |b|^2$, so $|b|^2 = 0$ and $b = 0$. ($\Leftarrow$) A diagonal matrix is normal, since $\mathbf{D}^*\mathbf{D} = \mathbf{D}\mathbf{D}^* = \operatorname{diag}(|a|^2, |d|^2)$.

(c) A unitary matrix is normal, so by (b) it must be diagonal, $\operatorname{diag}(a, d)$, and then $\mathbf{A}^*\mathbf{A} = \operatorname{diag}(|a|^2, |d|^2) = \mathbf{I}$ forces $|a| = |d| = 1$. Conversely every such matrix is unitary. So the upper triangular unitary matrices are exactly $\operatorname{diag}(a, d)$ with $|a| = |d| = 1$.

12.3.4.3 C. Going deeper

Exercise. C1: A normal operator with no proper invariant subspace

Let V be a finite-dimensional complex inner product space with $V \neq \{0\}$, and let $T \in \mathcal{L}(V)$ be normal. Suppose the only T-invariant subspaces of V are $\{0\}$ and V.

(a) Show that $\dim V = 1$.

(b) Show that the conclusion fails over $\mathbb{R}$, by exhibiting a normal operator on $\mathbb{R}^2$ whose only invariant subspaces are $\{\mathbf{0}\}$ and $\mathbb{R}^2$.

(c) Where exactly does the argument of (a) use that the field is $\mathbb{C}$?

Hint: an eigenvector spans an invariant subspace.

Solution.

(a) Since $V \neq \{0\}$ and the field is $\mathbb{C}$, T has an eigenvector $v \neq 0$ (Theorem 9.2.4). Then $\operatorname{span}(v)$ is T-invariant and nonzero, so by hypothesis $\operatorname{span}(v) = V$ and $\dim V = 1$. Normality is not needed for this direction.

(b) The quarter turn $\mathbf{R}_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is orthogonal, hence normal. A 1-dimensional invariant subspace would be spanned by a real eigenvector, and $\mathbf{R}_{\pi/2}$ has none: its characteristic polynomial is $x^2 + 1$, with no real root. So its only invariant subspaces are $\{\mathbf{0}\}$ and $\mathbb{R}^2$, while $\dim \mathbb{R}^2 = 2$.

(c) Only in the existence of an eigenvector. Over $\mathbb{C}$ that is the Fundamental Theorem of Algebra (Theorem 6.5.1) reaching the characteristic polynomial; over $\mathbb{R}$ the polynomial may have no root, which is exactly what (b) exploits. Note that Proposition 12.3.6 still holds over $\mathbb{R}$ — it says nothing about invariant subspaces *existing*.

Exercise. C2: Arithmetic of normal operators

Let $S, T \in \mathcal{L}(V)$ be normal.

(a) Prove that T^* is normal and that cT is normal for every $c \in F$.

(b) Prove that $p(T)$ is normal for every polynomial $p \in F[x]$.

(c) Give a concrete S and T, both normal, with ST not normal. (Part (b) shows that this cannot happen when S is a polynomial in T.)

Solution.

(a) $(T^*)^*T^* = TT^* = T^*T = T^*(T^*)^*$, using $T^{**} = T$ (Theorem 11.6.2 (d)). For cT, by Theorem 11.6.2 (b) we have $(cT)^* = \overline{c}T^*$, so

$$(cT)^*(cT) = \overline{c}c\,T^*T = |c|^2\,TT^* = (cT)(cT)^*.$$

(b) Write $p(x) = \sum_k a_k x^k$. Then $p(T)^* = \sum_k \overline{a_k}(T^*)^k$ by Theorem 11.6.2 (a), (b), and (c). Every power of T commutes with every power of T^*: from $TT^* = T^*T$ one moves a T^* past one T at a time. Hence $p(T)$ and $p(T)^*$, being sums of products of such powers, commute, and $p(T)$ is normal.

(c) Take $S = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, both symmetric and hence normal. Then

$$ST = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix},$$

$$(ST)^\top(ST) = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \qquad (ST)(ST)^\top = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix},$$

so ST is not normal.

Exercise. C3: Normality depends on the inner product

Normality is a property of T **together with** an inner product, not of T alone. Work on $V = \mathbb{R}^2$ with the operator T whose standard matrix is $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

(a) Verify that T is normal for the standard inner product.

(b) Now use $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{G}} = \mathbf{y}^\top \mathbf{G}\mathbf{x}$ with $\mathbf{G} = \operatorname{diag}(1, 2)$, which is an inner product on $\mathbb{R}^2$. Using the rule $[T^*]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{G}^{-1}\mathbf{A}^\top\mathbf{G}$ in the standard basis $\mathcal{E}$ (Exercise (C3: The adjoint in an arbitrary basis)), compute the matrix of the adjoint of T for this inner product.

(c) Show that T is **not** normal for $\langle \cdot, \cdot \rangle_{\mathbf{G}}$, and reconcile this with (a).

Solution.

(a) $\mathbf{A}^\top\mathbf{A} = \mathbf{A}\mathbf{A}^\top = \mathbf{I}$, so T is even orthogonal, hence normal.

(b) $\mathbf{G}^{-1} = \operatorname{diag}(1, \frac{1}{2})$, so

$$\mathbf{G}^{-1}\mathbf{A}^\top\mathbf{G} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2 \\ -\frac{1}{2} & 0 \end{pmatrix}.$$

Call this matrix $\mathbf{A}^\star$.

(c) Computing both products,

$$\mathbf{A}\mathbf{A}^{\star} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}, \qquad \mathbf{A}^{\star}\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix},$$

which differ, so T is not normal for $\langle \cdot, \cdot \rangle_{\mathbf{G}}$. There is no contradiction: the adjoint is built from the inner product (Definition 11.6.1), so changing the inner product changes T^*, and with it the equation $T^*T = TT^*$. The standard basis is orthonormal for the first inner product and not for the second, which is exactly the hypothesis Theorem 11.6.3 needs.